Counter Numeric Display

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Intro to Electrical Engineering

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Lab Report #4

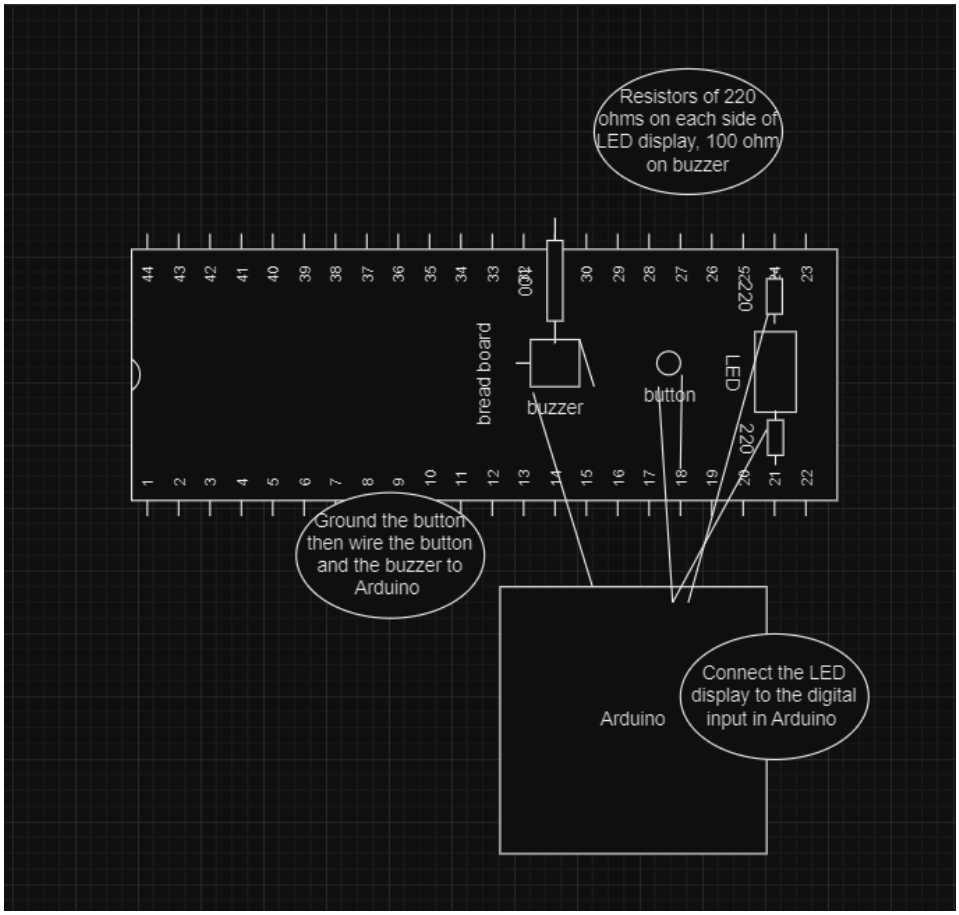
Overview

I'm going to have an LED segment display that displays a number, a buzzer, and a button. When I click the button the number should count from 0 to 9, and each time the number switches the buzzer should make a sound. If you let go of the button it should finish counting or if you don't click on the button the decimal place should flash three times and make a buzzer noise.

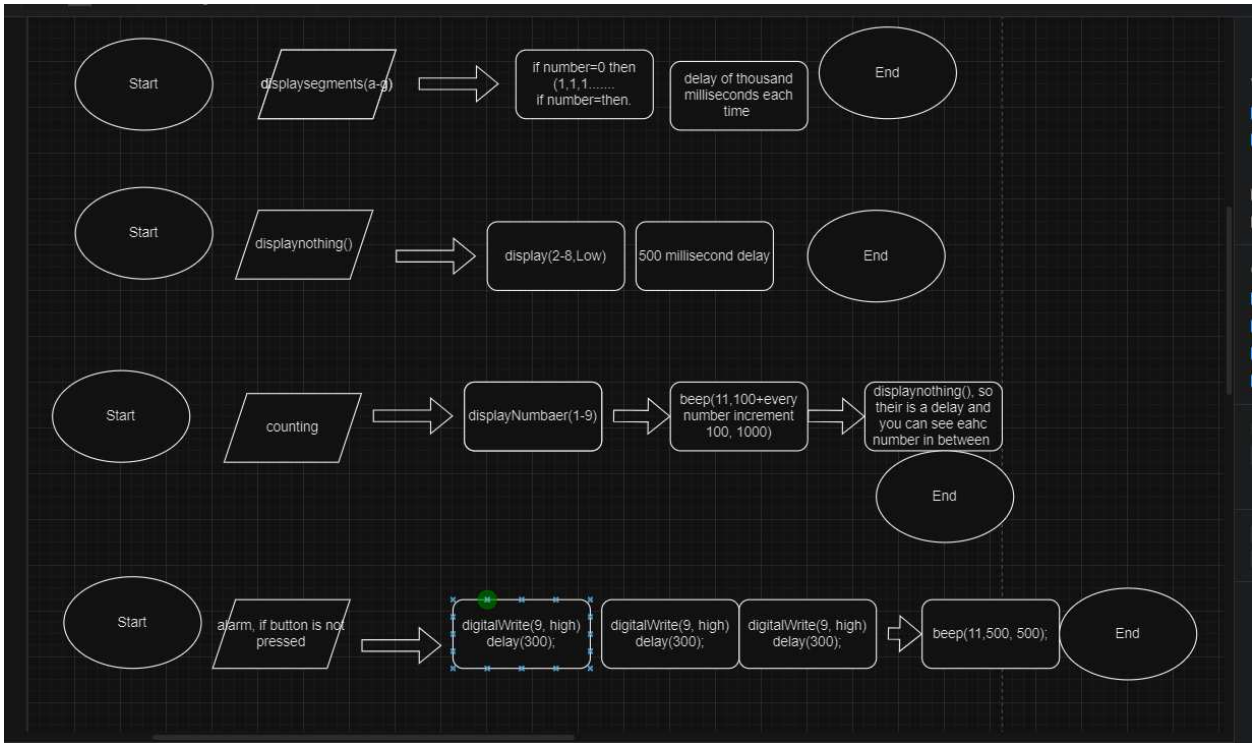
Requirements

The LED segment display is going to be broken down into 8 different segments. Each segment is going to light up to resemble a number. I'm going to do this by using the displaySegment() code and inputting 1 to light up and 0 to shut off. I'm going to light up the digital input 9 since that's the decimal point, and make it light up 3 times each with a 300-millisecond delay, and then the buzzer noise at 500 after, and this is for when the button is not being clicked. The buzzer noise is also going to be activated when the number is incrementing up counting from 0 to 9, and every time the number changes the buzzer is going to get louder by an increment of 100 starting at 100 at 0.

Wire Diagram



Software Diagram



Design Verification

https://youtube.com/shorts/yMfSnW0xKaU - Video Explaining Project

sketch_oct23a.ino

```
void setup() {
pinMode(10, INPUT_PULLUP);
}
void Beep(int BuzzPin, int Hz, int time)
{
  tone(BuzzPin, Hz, time);
}

void displaySegments(int a, int b, int c, int d, int e, int f, int g)
    {
  digitalWrite(2, a);
  digitalWrite(3, b);
  digitalWrite(4, c);
  digitalWrite(5, d);
  digitalWrite(6, e);
  digitalWrite(7, f);
  digitalWrite(8, g);
  delay(500);
    }

void displayNumber(int number)
    {

      if(number == 0)
      {
        displaySegments(1, 1, 1, 1, 1, 1, 0);
        delay(1000);
      }
      if(number == 1)
      {
        displaySegments(0, 1, 1, 0, 0, 0, 0);
        delay(1000);
      }
```

sketch_oct23a.ino

```
    delay(1000);
  }
  if(number == 2)
  {
    displaySegments(1, 1, 0, 1, 1, 0, 1);
    delay(1000);
  }
  if(number == 3)
  {
    displaySegments(1, 1, 1, 1, 0, 0, 1);
    delay(1000);
  }
  if(number == 4)
  {
    displaySegments(0, 1, 1, 0, 0, 1, 1);
    delay(1000);
  }
  if(number == 5)
  {
    displaySegments(1, 0, 1, 1, 0, 1, 1);
    delay(1000);
  }
  if(number == 6)
  {
    displaySegments(1, 0, 1, 1, 1, 1, 1);
    delay(1000);
  }
  if(number == 7)
  {
    displaySegments(1, 1, 1, 0, 0, 0, 0);
    delay(1000);
  }
  if(number == 8)
```

Output

```
      }
      if(number == 8)
      {
        displaySegments(1, 1, 1, 1, 1, 1, 1);
        delay(1000);
      }
      if(number == 9)
      {
        displaySegments(1, 1, 1, 0, 0, 1, 1);
        delay(1000);
      }
    }
void displayNothing() //Function that displays nothing
    {
    digitalWrite(2, LOW);
    digitalWrite(3, LOW);
    digitalWrite(4, LOW);
    digitalWrite(5, LOW);
    digitalWrite(6, LOW);
    digitalWrite(7, LOW);
    digitalWrite(8, LOW);
    delay(500);
    }
void count()

    {
      displayNumber(0);
      Beep(11, 150, 1000);
      displayNothing();
      displayNumber(1);
      Beep(11, 200, 1000);
      displayNothing();
```

```
    Beep(11, 200, 1000);
    displayNothing();
    displayNumber(2);
    Beep(11, 300, 1000);
    displayNothing();
    displayNumber(3);
    Beep(11, 400, 1000);
    displayNothing();
    displayNumber(4);
    Beep(11, 500, 1000);
    displayNothing();
    displayNumber(5);
    Beep(11, 600, 1000);
    displayNothing();
    displayNumber(6);
    Beep(11, 700, 1000);
    displayNothing();
    displayNumber(7);
    Beep(11, 800, 1000);
    displayNothing();
    displayNumber(8);
    Beep(11, 900, 1000);
    displayNothing();
    displayNumber(9);
    Beep(11, 1000, 1000);
    displayNothing();
  }

void alarm() //decimal spot 3 times and beep if button not pressed
  {
    digitalWrite(9, HIGH);
    delay(300);
    digitalWrite(9, LOW);
```

out

```
    Beep(11, 1000, 1000);
    displayNothing();
  }

void alarm() //decimal spot 3 times and beep if button not pressed
  {
    digitalWrite(9, HIGH);
    delay(300);
    digitalWrite(9, LOW);
    delay(300);
    digitalWrite(9, HIGH);
    delay(300);
    digitalWrite(9, LOW);
    delay(300);
    digitalWrite(9, HIGH);
    delay(300);
    digitalWrite(9, LOW);
    delay(300);
    Beep(11, 500, 500);
  }

void loop()

{
if(digitalRead(10) == LOW)
  {
    count();
  }
else
  {
    alarm();
  }
}
```

Learnings

I learned a lot about for loops as well as if and else statements.How they can be applied to projects and really expand what you can do. I also learned more about the coding binary of 1 and 0 through the “display segment” code. The wiring also refreshed and improved my knowledge on grounding.

Conclusions

I was successful in that our project worked and fulfilled what was required. But where I could've been better was using less lines of code and maybe figuring out a way to incorporate more loops into my project so I could have done it more efficiently perhaps. But overall I would see it as a success since it worked and you could actually see numbers displayed on my project.